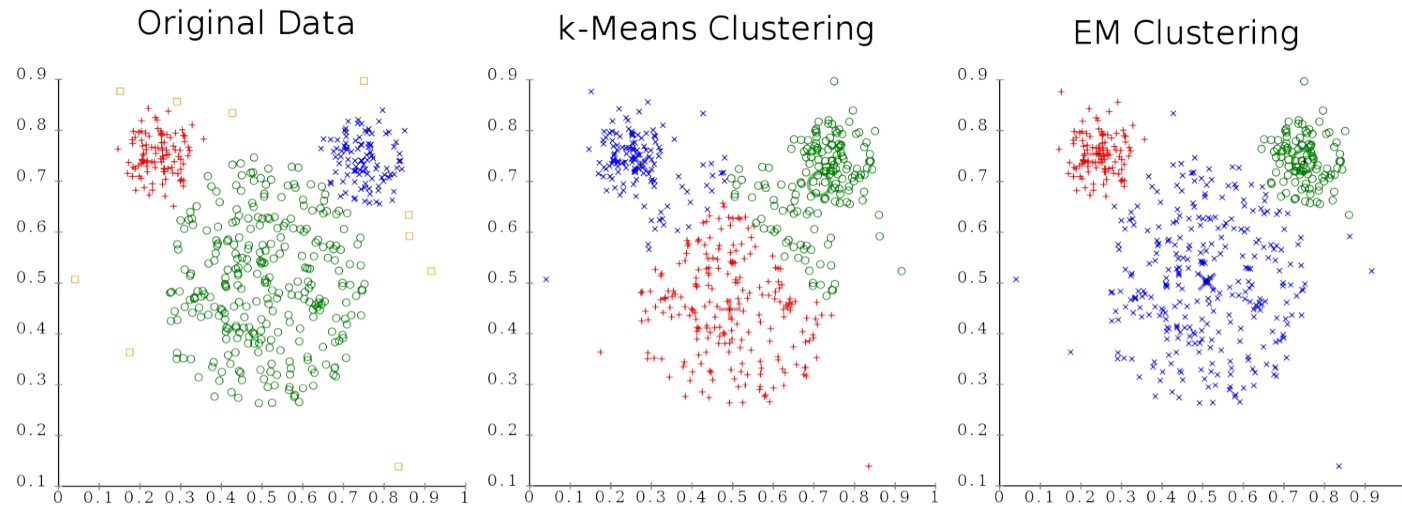


Lecture 16 - The Expectation- Maximization Algorithm

In some settings, the k-means algorithm does not give desirable results:

Different cluster analysis results on "mouse" data set:



The issue is that k-means does not account for the size of the different clusters.

Alternative approach: assume x_i 's are drawn at random from a mixture of Gaussians distribution and cluster using the Expectation Maximization (EM) algorithm.

A Gaussian distribution is a probability distribution that characterizes how likely different values of a random variable are.

$$f(\underline{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\} \quad (\text{shorthand } \mathcal{N}(\underline{\mu}, \Sigma))$$

$\underline{\mu}$ = mean = $\mathbb{E} \underline{x}$ = expected (average) value of $\underline{x}$

Σ = covariance = $\mathbb{E} [(\underline{x} - \underline{\mu})(\underline{x} - \underline{\mu})^T] \Leftrightarrow \Sigma_{ij} = \mathbb{E} [(x_i - \mu_i)(x_j - \mu_j)]$

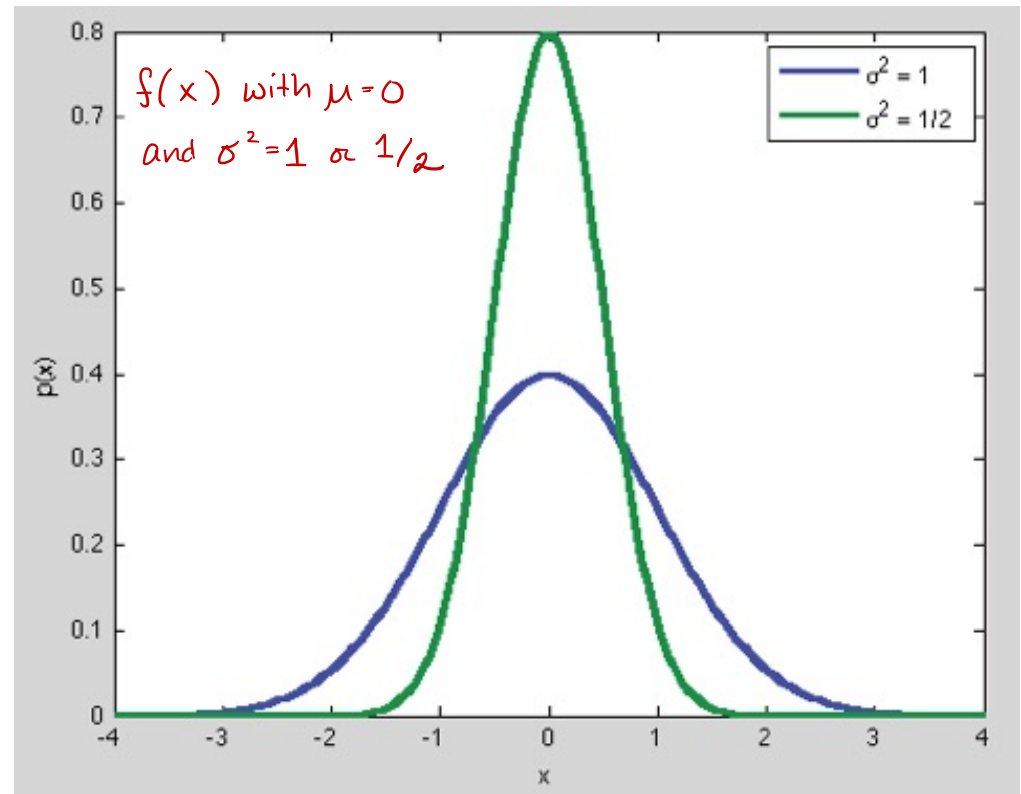
$|\Sigma|$ = matrix determinant
= product of singular values

Ex 1: $p=1$

$x \sim \mathcal{N}(\mu, \sigma^2)$

(i.e. $\Sigma = \sigma^2$)

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

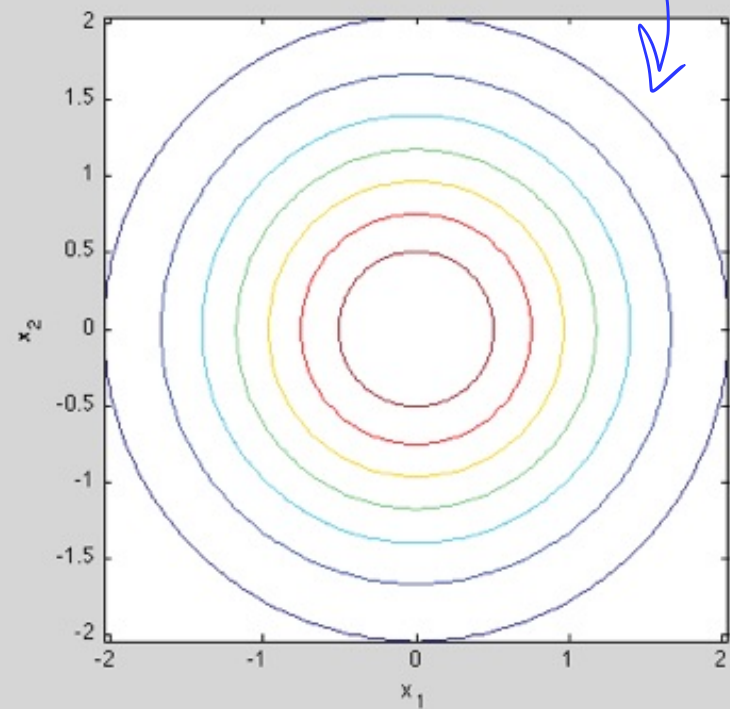
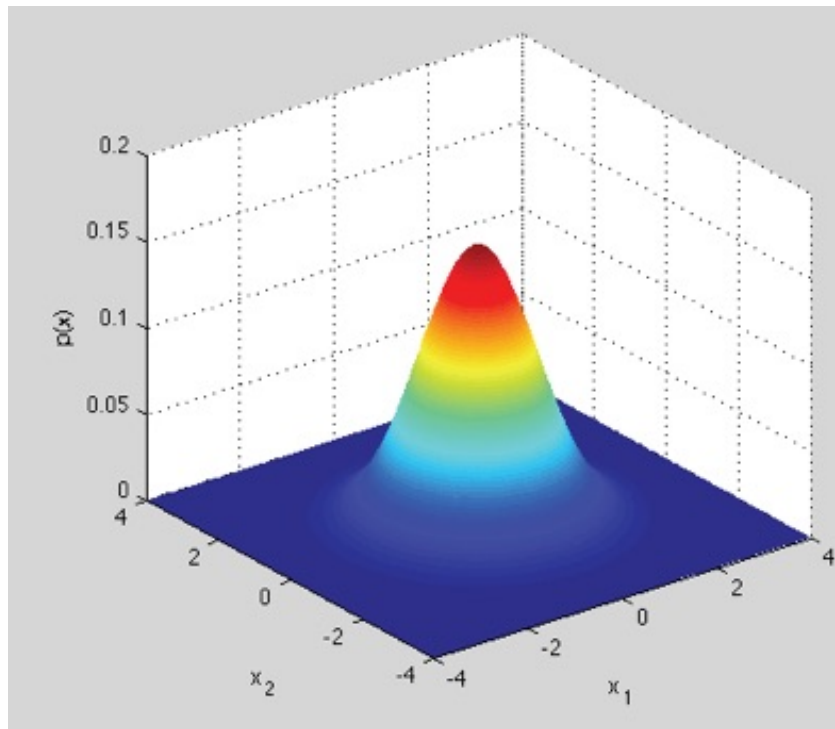


Ex. $p = 2$

$$\underline{x} \sim \mathcal{N}(\underline{\mu}, \sigma^2 \mathbf{I})$$
$$\hookrightarrow \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix}$$

$$f(\underline{x}) = \gamma \iff \frac{(x_1 - \mu_1)^2}{\sigma^2} + \frac{(x_2 - \mu_2)^2}{\sigma^2} = \frac{\|\underline{x} - \underline{\mu}\|_2^2}{\sigma^2} = \gamma'$$

Contour plot. each circle is set of all $\underline{x}$
such that $f(\underline{x}) = \gamma$ for some γ
= set of $\underline{x}$ such that $\|\underline{x} - \underline{\mu}\|^2 = \gamma'$



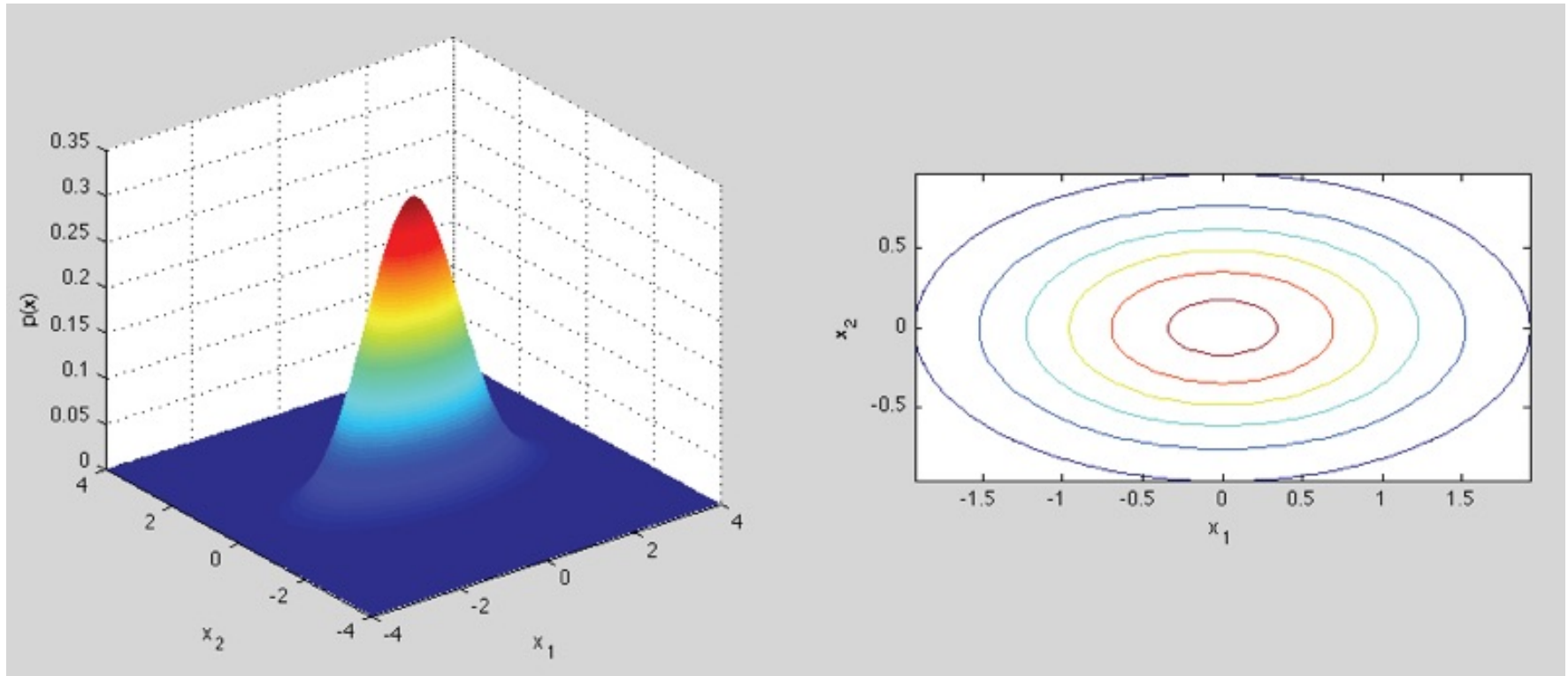
Ex. $p=2$.

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$\underline{x} \sim \mathcal{N}(\underline{\mu}, \Sigma)$$

Here density contours are ellipses whose axes align with the coordinate axes. Note:

$$f(\underline{x}) = \gamma \Leftrightarrow \frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} = \gamma'$$



Ex. Σ is an arbitrary positive-definite matrix.

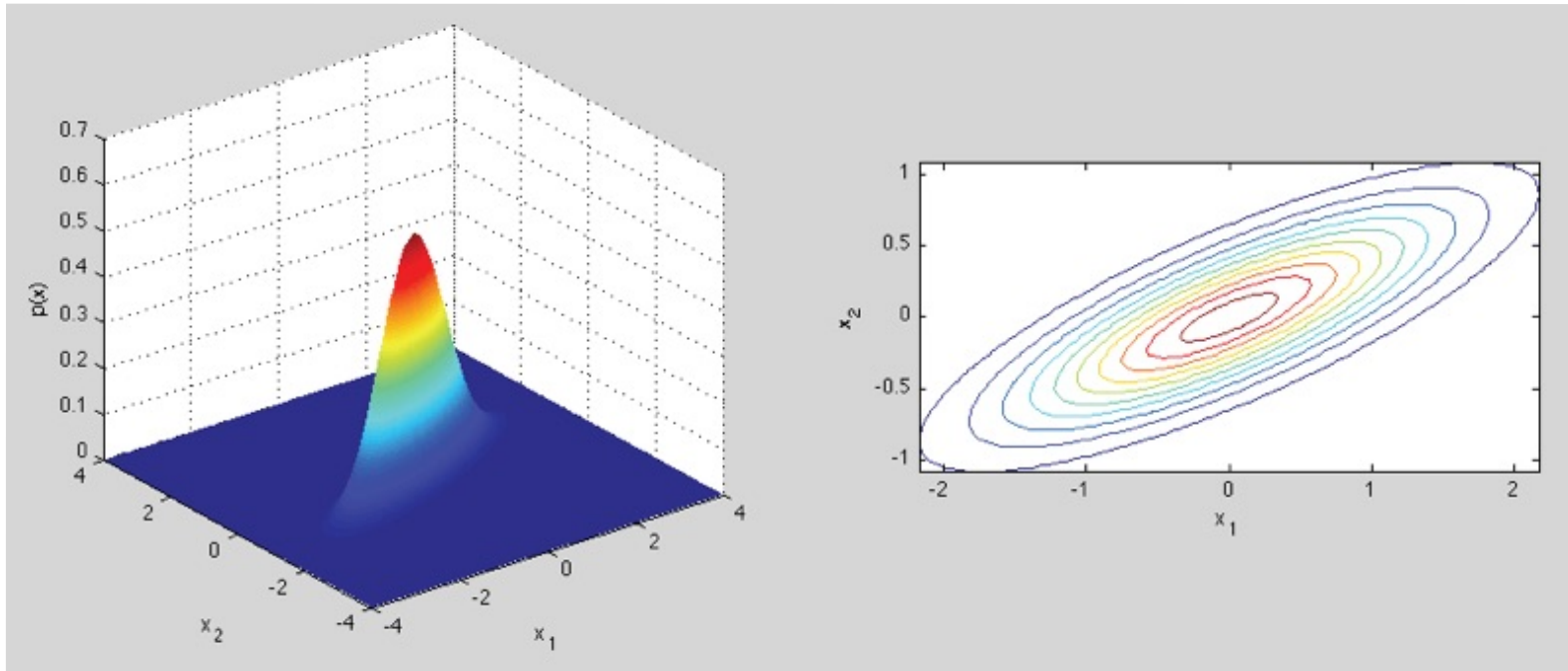
Let $\Sigma = V\Lambda V^T$ (eigenvalue decomposition)

Let $\underline{x}' = V^T \underline{x}$ and $\underline{\mu}' = V^T \underline{\mu}$. $\Leftarrow \underline{x}' = \underline{x}$ in rotated coordinate system defined by columns of V .

Then contour = set of all $\underline{x}$ s.t. $(\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = \gamma$ for some γ

$$(\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = (\underline{x} - \underline{\mu})^T V \Lambda^{-1} V^T (\underline{x} - \underline{\mu}) = (\underline{x}' - \underline{\mu}')^T \Lambda^{-1} (\underline{x}' - \underline{\mu}') = \frac{(x'_1 - \mu'_1)^2}{\lambda_1} + \frac{(x'_2 - \mu'_2)^2}{\lambda_2}$$

= ellipse in rotated coordinate system, where V defines rotation.



The SVD of the covariance matrix Σ tells us how the distribution is "oriented" and spread out in different directions

Maximum likelihood estimation

Given $\underline{x}_i \sim \mathcal{N}(\underline{\mu}, \underline{\Sigma})$ for $i=1, 2, \dots, n$, we may wish to estimate $\underline{\mu}$ and $\underline{\Sigma}$.

Maximum likelihood strategy: Choose $\hat{\underline{\mu}}$ and $\hat{\underline{\Sigma}}$ to maximize the likelihood of the $\underline{x}_i$'s

$$\begin{aligned}(\hat{\underline{\mu}}, \hat{\underline{\Sigma}}) &= \underset{\underline{\mu}, \underline{\Sigma}}{\operatorname{argmax}} \prod_{i=1}^n f(\underline{x}_i; \underline{\mu}, \underline{\Sigma}) \\&= \underset{\underline{\mu}, \underline{\Sigma}}{\operatorname{argmax}} \log \left(\prod_{i=1}^n f(\underline{x}_i; \underline{\mu}, \underline{\Sigma}) \right) \\&= \underset{\underline{\mu}, \underline{\Sigma}}{\operatorname{argmin}} -\log \left(\prod_{i=1}^n f(\underline{x}_i; \underline{\mu}, \underline{\Sigma}) \right)\end{aligned}$$

$$\begin{aligned}&= \underset{\underline{\mu}, \underline{\Sigma}}{\operatorname{argmin}} \sum_{i=1}^n -\log f(\underline{x}_i; \underline{\mu}, \underline{\Sigma}) \\&= \underset{\underline{\mu}, \underline{\Sigma}}{\operatorname{argmin}} \sum_{i=1}^n \frac{1}{2} \log(2\pi) - \frac{1}{2} \log |\underline{\Sigma}| - \frac{1}{2} (\underline{x}_i - \underline{\mu})^T \underline{\Sigma}^{-1} (\underline{x}_i - \underline{\mu})\end{aligned}$$

Compute gradients, set to zero $\Rightarrow$

$$\hat{\underline{\mu}} = \frac{1}{n} \sum_{i=1}^n \underline{x}_i, \quad \hat{\underline{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\underline{x}_i - \hat{\underline{\mu}})(\underline{x}_i - \hat{\underline{\mu}})^T$$

Back to clustering

Assume there are K Gaussians (each will correspond to a different cluster) with means $\underline{\mu}_k$ and covariances Σ_k for $k=1, 2, \dots, K$.

We model the observed $\underline{x}_i$'s drawn from a mixture of these Gaussians as follows.

- choose one of the K Gaussians — i.e. the k^{th} Gaussian is chosen with probability π_k ,

where $\sum_{k=1}^K \pi_k = 1$; denote k_i

- draw $\underline{x}_i \sim \mathcal{N}(\underline{\mu}_{k_i}, \Sigma_{k_i})$

$$\Rightarrow f(\underline{x}_i) = \sum_{k=1}^K \pi_k f(\underline{x}_i; \underline{\mu}_k, \Sigma_k)$$

Given $\underline{x}_i$'s, we want to cluster them without

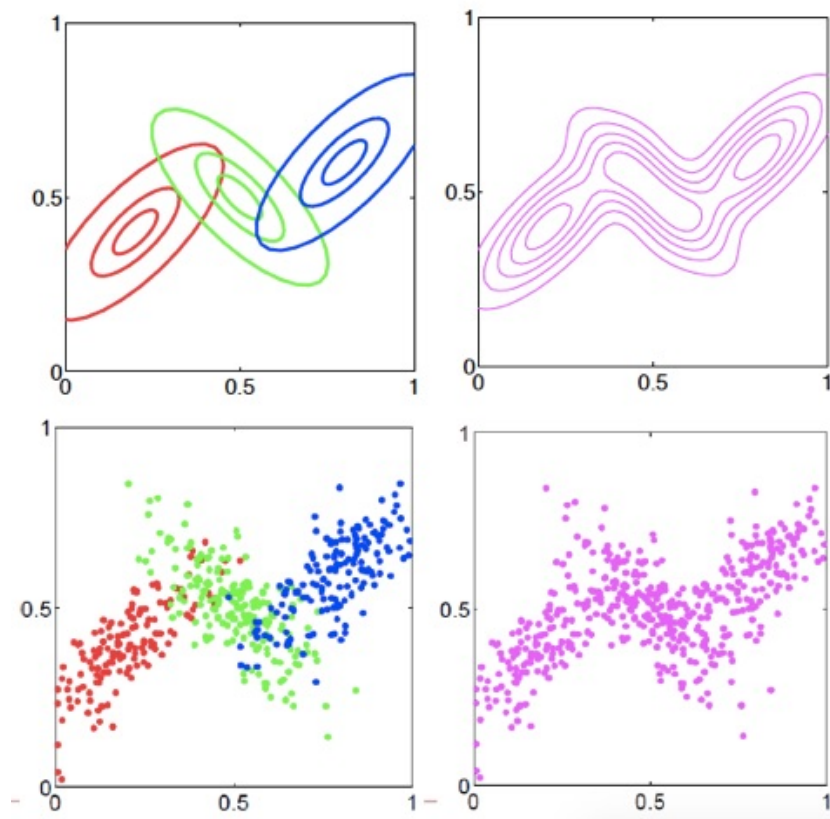
knowing $\underline{\mu}_k$'s or Σ_k 's or π_k 's.

if we knew cluster membership k_i for each $\underline{x}_i$,

then we could use maximum likelihood estimation to

compute $\underline{\mu}_k$'s, Σ_k 's, and π_k 's. Without k_i 's,

maximum likelihood estimation is hard.



Expectation - Maximization Algorithm

- Initialize means $\hat{\mu}_k$, covariances $\hat{\Sigma}_k$, and mixture weights $\hat{\pi}_k$ for $k=1, 2, \dots, K$
- **E-step:** Compute $p_k(\underline{x}_i)$ = Probability that $\underline{x}_i$ was drawn from k^{th} Gaussian given value of $\underline{x}_i$
$$= \Pr(k_i = k | \underline{x}_i) = \frac{\Pr(k_i = k) f(\underline{x}_i | k_i = k)}{f(\underline{x}_i)} \quad (\text{Bayes rule})$$

$$\hat{p}_k(\underline{x}_i) = \frac{\hat{\pi}_k f(\underline{x}_i | \hat{\mu}_k, \hat{\Sigma}_k)}{\sum_{j=1}^K \hat{\pi}_j f(\underline{x}_i | \hat{\mu}_j, \hat{\Sigma}_j)} \quad \text{for } k=1, 2, \dots, K$$

$i = 1, 2, \dots, n$

- **M-step:** using $\hat{p}_k(\underline{x}_i)$'s, update estimates of $\hat{\mu}_k$, $\hat{\Sigma}_k$, $\hat{\pi}_k$:

$$\hat{\mu}_k = \frac{\sum_{i=1}^n \hat{p}_k(\underline{x}_i) \underline{x}_i}{\sum_{i=1}^n \hat{p}_k(\underline{x}_i)}$$

$$\hat{\Sigma}_k = \frac{\sum_{i=1}^n \hat{p}_k(\underline{x}_i) (\underline{x}_i - \hat{\mu}_k)(\underline{x}_i - \hat{\mu}_k)^T}{\sum_{i=1}^n \hat{p}_k(\underline{x}_i)}$$

$$\hat{\pi}_k = \frac{1}{n} \sum_{i=1}^n \hat{p}_k(\underline{x}_i)$$

- if not converged, return to E-step.

